July 7, 2026

To the Editors of *Physical Review Letters*,

We are pleased to submit our manuscript, **"A temporally-constrained latent space reveals the stochastic dynamics of convective organization,"** by Sophie Abramian, Olivier Pauluis, and Pierre Gentine, for consideration as a Letter.

Many systems are both high-dimensional and stochastic, meaning that the same conditions can lead to different trajectories. A central goal is to reduce their dimension in order to better understand the underlying dynamics. In our study, we look for a reduction chosen on purpose to *simplify* the dynamics.

We introduce a framework that learns the reduction and the dynamics jointly: an encoder maps the system into a low-dimensional latent space, the latent space is partitioned into a discrete set of states, and transitions between states are modelled as a Markov chain. It builds on data-driven Koopman methods (Brunton and colleagues), which learn a latent space in which the dynamics become linear, and extends them from deterministic to *stochastic* evolution. The resulting description is compact, interpretable, and general—it applies to any complex system, from turbulence to neural activity.

We demonstrate it on a hard and important problem: the organization of atmospheric deep convection, one of the largest uncertainties in climate projections. We study it in a controlled, idealized setting—simulations of tropical squall lines under a fixed environment—so that the variability we model is *internal* to the convection itself. With no supervision, the method recovers five physically meaningful convective states and a Markov chain that alternates between organized and disorganized periods. We also assess how well the Markov description holds, and find that *where it breaks down, it points to a hidden variable the model is missing.* This leads us to cold pools—masses of cold air beneath storms—which we identify as a measurable source of that memory, accounting for about a third of the gap.

We believe the work fits *PRL* for three reasons. First, it offers a general, data-driven way to discover the states of a complex system and model the transitions between them—of broad interest across physics. Second, the same Markov description doubles as a discovery tool: where the transitions stop being memoryless, they reveal a hidden variable the model is missing. Third, the problem it tackles, the internal variability of convection, matters for climate.

The manuscript has not been published and is not under consideration elsewhere; all authors have approved this submission.

Suggested reviewers: Zhiming Kuang, Harvard University (kuang@fas.harvard.edu); Allison Wing, Florida State University (awing@fsu.edu); Cathy Hohenegger, MPI for Meteorology (cathy.hohenegger@mpimet.mpg.de); Brian Mapes, University of Miami (bmapes@miami.edu); Elizabeth Barnes, Colorado State University (eabarnes@colostate.edu).

Thank you for considering our manuscript.

Sincerely,



Sophie Abramian, on behalf of the authors
Department of Earth and Environmental Engineering, Columbia University
sophie.abramian@gmail.com